RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)
```

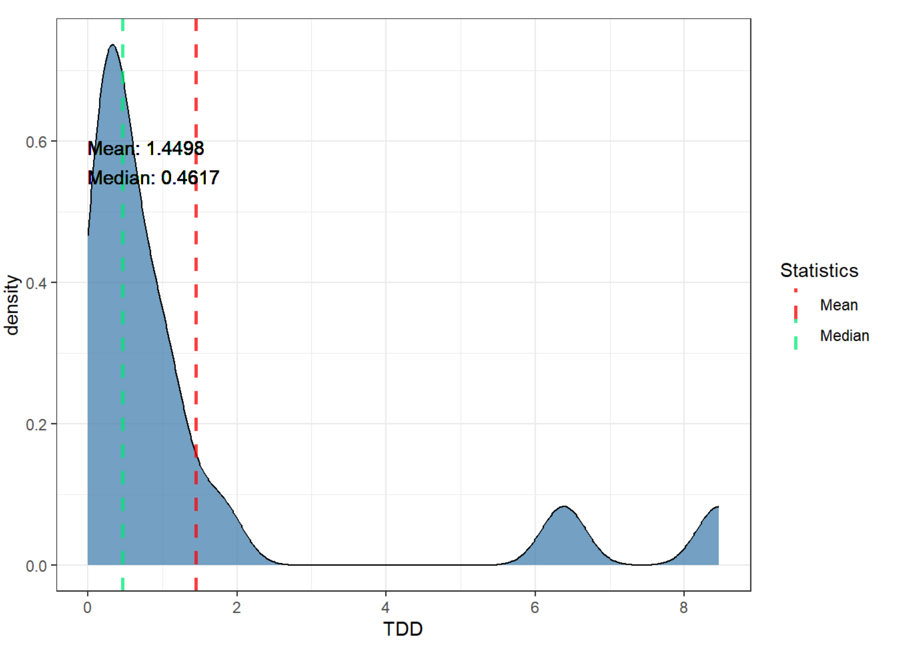
```
## Rows: 220
## Columns: 11
## $ repo              <chr> "wicket", "wicket", "wicket", "wicket", "wicket", …
## $ pr_number         <int> 385, 385, 385, 385, 385, 385, 456, 456, 456, 456, …
## $ rule              <chr> "java:S1124", "java:S1068", "java:S1948", "java:S1…
## $ severity          <chr> "MINOR", "MAJOR", "CRITICAL", "MAJOR", "MAJOR", "M…
## $ file              <chr> "wicket-core/src/test/java/org/apache/wicket/core/…
## $ type              <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status            <chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O…
## $ debt              <int> 2, 5, 30, 5, 5, 1, 8, 8, 8, 10, 10, 8, 12, 8, 8, 8…
## $ ncloc_affected_file <int> 40, 40, 40, 40, 40, 40, 394, 394, 394, 394, 394, 3…
## $ complexity        <int> 5, 5, 5, 5, 5, 5, 39, 39, 39, 39, 39, 39, 39, 39, …
## $ origin            <chr> "NEW", "NEW", "NEW", "NEW", "NEW", "NEW", "PRE-EXI…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 16 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 16 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

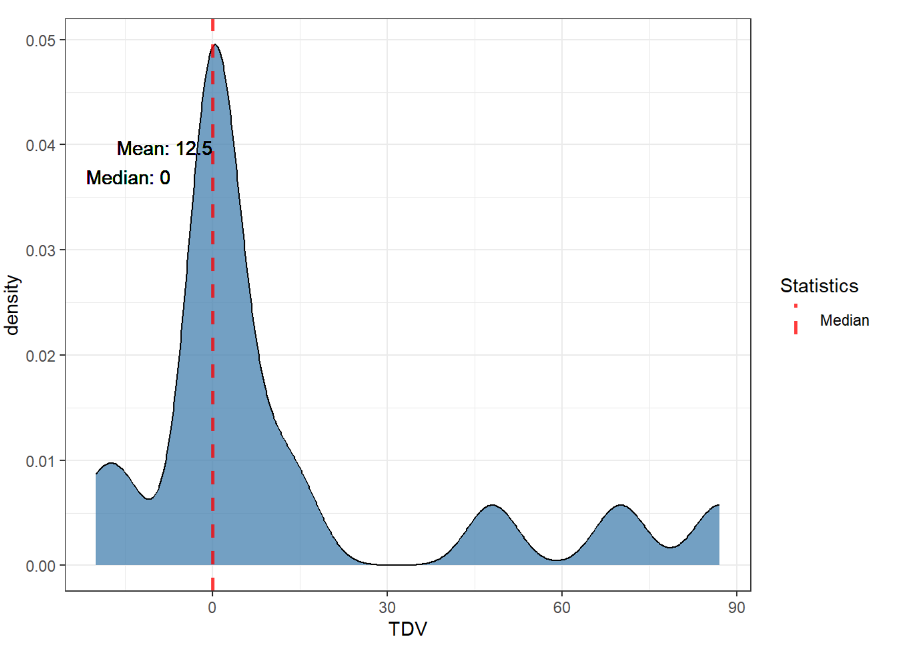


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 16 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 16 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 × 4
##   repo   tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>                <dbl>             <dbl>                 <dbl>
## 1 wicket                12.5                50                  37.5
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                      <dbl>
## 1                     12.5                     50                       37.5
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 × 3
##   repo   preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>                          <dbl>                             <dbl>
## 1 wicket                          61.5                              38.5
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                          61.5                                   38.5
```

Percentages for new TDV by repo

```
## # A tibble: 1 × 3
## # Groups:   repo [1]
##   repo   new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                  <dbl>                     <dbl>
## 1 wicket                    90                        10
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                         90                             10
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositiries.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank wicket
##    <int> <chr>
##  1     1 java:S112  -  4
##  2     2 java:S1192  -  3
##  3     3 java:S1874  -  2
##  4     4 java:S3398  -  2
##  5     5 java:S110  -  2
##  6     6 java:S5542  -  2
##  7     7 java:S3329  -  2
##  8     8 java:S1319  -  2
##  9     9 java:S4042  -  1
## 10    10 java:S1155  -  1
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S112             122
##  2 java:S1192            123
##  3 java:S110             124
##  4 java:S1319            125
##  5 java:S1874            126
##  6 java:S3329            127
##  7 java:S3398            128
##  8 java:S5542            129
##  9 java:S1155            130
## 10 java:S3358            131
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank wicket
##    <int> <chr>
##  1     1 java:S1192  -  35
##  2     2 java:S1602  -  29
##  3     3 java:S100  -  14
##  4     4 java:S125  -  13
##  5     5 java:S110  -  13
##  6     6 java:S1125  -  10
##  7     7 java:S112  -  9
##  8     8 java:S3776  -  4
##  9     9 java:S1604  -  4
## 10    10 java:S1186  -  4
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S1192            122
##  2 java:S1602            123
##  3 java:S100             124
##  4 java:S110             125
##  5 java:S125             126
##  6 java:S1125            127
##  7 java:S112             128
##  8 java:S1186            129
##  9 java:S1604            130
## 10 java:S2479            131
```

Outliers

Higher TDD

```
## # A tibble: 16 × 5
##    repo   pr_number total_debt total_ncloc    tdd
##    <chr>      <int>      <int>       <int>  <dbl>
##  1 wicket       709       2218         262 8.47
##  2 wicket       708       1161         182 6.38
##  3 wicket       462       1103         619 1.78
##  4 wicket       385         48          40 1.2
##  5 wicket       607        245         241 1.02
##  6 wicket       587         87          97 0.897
##  7 wicket       657         95         117 0.812
##  8 wicket       605         44          81 0.543
##  9 wicket       684         81         213 0.380
## 10 wicket       598         15          45 0.333
## 11 wicket       456        119         394 0.302
## 12 wicket       601        130         447 0.291
## 13 wicket       761        115         414 0.278
## 14 wicket       794        409        1703 0.240
## 15 wicket       621         10          54 0.185
## 16 wicket       798         15         162 0.0926
```

Higher TDV

```
## # A tibble: 16 × 3
##    pr_number repo     tdv
##        <int> <chr>  <int>
##  1       587 wicket    87
##  2       607 wicket    70
##  3       385 wicket    48
##  4       598 wicket    15
##  5       761 wicket    10
##  6       684 wicket     5
##  7       456 wicket     0
##  8       601 wicket     0
##  9       605 wicket     0
## 10       621 wicket     0
## 11       657 wicket     0
## 12       708 wicket     0
## 13       794 wicket     0
## 14       798 wicket     0
## 15       462 wicket   -15
## 16       709 wicket   -20
```

Lower TDV

```
## # A tibble: 16 × 3
##    pr_number repo     tdv
##        <int> <chr>  <int>
##  1       709 wicket   -20
##  2       462 wicket   -15
##  3       456 wicket     0
##  4       601 wicket     0
##  5       605 wicket     0
##  6       621 wicket     0
##  7       657 wicket     0
##  8       708 wicket     0
##  9       794 wicket     0
## 10       798 wicket     0
## 11       684 wicket     5
## 12       761 wicket    10
## 13       598 wicket    15
## 14       385 wicket    48
## 15       607 wicket    70
## 16       587 wicket    87
```

Extra

Number of PRs with pre-existing TD

```
##    n
## 1 13
```

Number of java:S1192 issues that affect test code

```
##    n
## 1 38
```

Number of java:S1192 issues that affect not test code

```
##   n
## 1 0
```

Summary NCLOC

```
##      ncloc
##  Min.   :  40.0
##  1st Qu.:  93.0
##  Median : 197.5
##  Mean   : 316.9
##  3rd Qu.: 399.0
##  Max.   :1703.0
```